SIMATS ENGINEERING

ASSESSMENT TOOL 1 - High (INDUSTRY PROBLEM SOLVING TASKS)

COURSE CODE: CSA09 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks: 50**

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASKS

Code of Conduct:

*I **NEELAKANDAMOORTHY C(192421063)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: NEELAKANDAMOORTHY C(192421063)

ANNEXURE

INDUSTRY PROBLEM SOLVING TASKS

5. Online Examination System Interface (BTL: BL5 – Evaluate)

Design a Swing-based GUI application for conducting online exams, including features like question navigation, timer display, and result submission. Use components such as radio buttons, panels, and menus. Implement event handling for answering questions and submitting responses. Validate user actions to prevent multiple submissions. Evaluate layout managers for organizing question panels and controls. Analyze usability and responsiveness under multiple users and propose enhancements for reliability and scalability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Industry Problem	**20%**	**Comprehensive understanding of real-world problem and constraints**	**Good understanding with minor gaps in requirements**	**Basic understanding; some requirements unclear**	**Poor understanding; misinterprets problem context**
Application of Engineering Concepts	**25%**	**Applies advanced and relevant concepts effectively to solve the problem**	**Applies appropriate concepts with minor errors**	**Limited application of concepts**	**Incorrect or no application of concepts**

Solution Design	**25%**	**Innovative, practical, and feasible solution aligned with industry standards**	**Logical and feasible solution with minor gaps**	**Basic solution with limited feasibility**	**Unclear or impractical solution**
Use of Tools	**15%**	**Effective use of modern tools, technologies, and real data**	**Appropriate use of tools with correct implementation**	**Limited or inconsistent use of tools**	**Minimal or incorrect use of tools**
Analysis	**10%**	**Thorough analysis with validated results and performance evaluation**	**Good analysis with reasonable validation**	**Basic analysis with limited validation**	**Poor or no analysis**
Professionalism	**5%**	**Highly professional, well-documented, and clearly presented work**	**Good documentation and presentation**	**Basic documentation with some gaps**	**Poor documentation and presentation**

JAVA CODE:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
public class OnlineExamApplet extends JApplet implements ActionListener {
    JLabel questionLabel, timerLabel, resultLabel;
    JRadioButton option1, option2, option3, option4;
    ButtonGroup options;
    JButton nextButton, submitButton;
    int question = 1
    int score = 0;
    int time = 60;
    boolean submitted = false;
    Timer timer;
    public void init() {
        setLayout(new BorderLayout())
        JPanel topPanel = new JPanel(new BorderLayout());
        JLabel title = new JLabel(
            "ONLINE EXAMINATION SYSTEM",
            JLabel.CENTER
        );
        title.setFont(new Font("Arial", Font.BOLD, 20));
        timerLabel = new JLabel("Time: 60 seconds");
        topPanel.add(title, BorderLayout.CENTER);
        topPanel.add(timerLabel, BorderLayout.EAST);
        add(topPanel, BorderLayout.NORTH);
        JPanel questionPanel = new JPanel();
        questionPanel.setLayout(new GridLayout(5, 1));
        questionLabel = new JLabel(
```

```
);
option1 = new JRadioButton("Python");
option2 = new JRadioButton("Java");
option3 = new JRadioButton("HTML");
option4 = new JRadioButton("CSS");
options = new ButtonGroup();
options.add(option1);
options.add(option2);
options.add(option3);
options.add(option4);
questionPanel.add(questionLabel);
questionPanel.add(option1);
questionPanel.add(option2);
questionPanel.add(option3);
questionPanel.add(option4);
add(questionPanel, BorderLayout.CENTER)
JPanel bottomPanel = new JPanel();
nextButton = new JButton("Next");
submitButton = new JButton("Submit");
resultLabel = new JLabel("");
nextButton.addActionListener(this);
submitButton.addActionListener(this);
bottomPanel.add(nextButton);
bottomPanel.add(submitButton);
bottomPanel.add(resultLabel);
add(bottomPanel, BorderLayou
JMenuBar menuBar = new JMenuBar();
JMenu examMenu = new JMenu("Exam");
JMenuItem startItem = new JMenuItem("Start Exam");
```

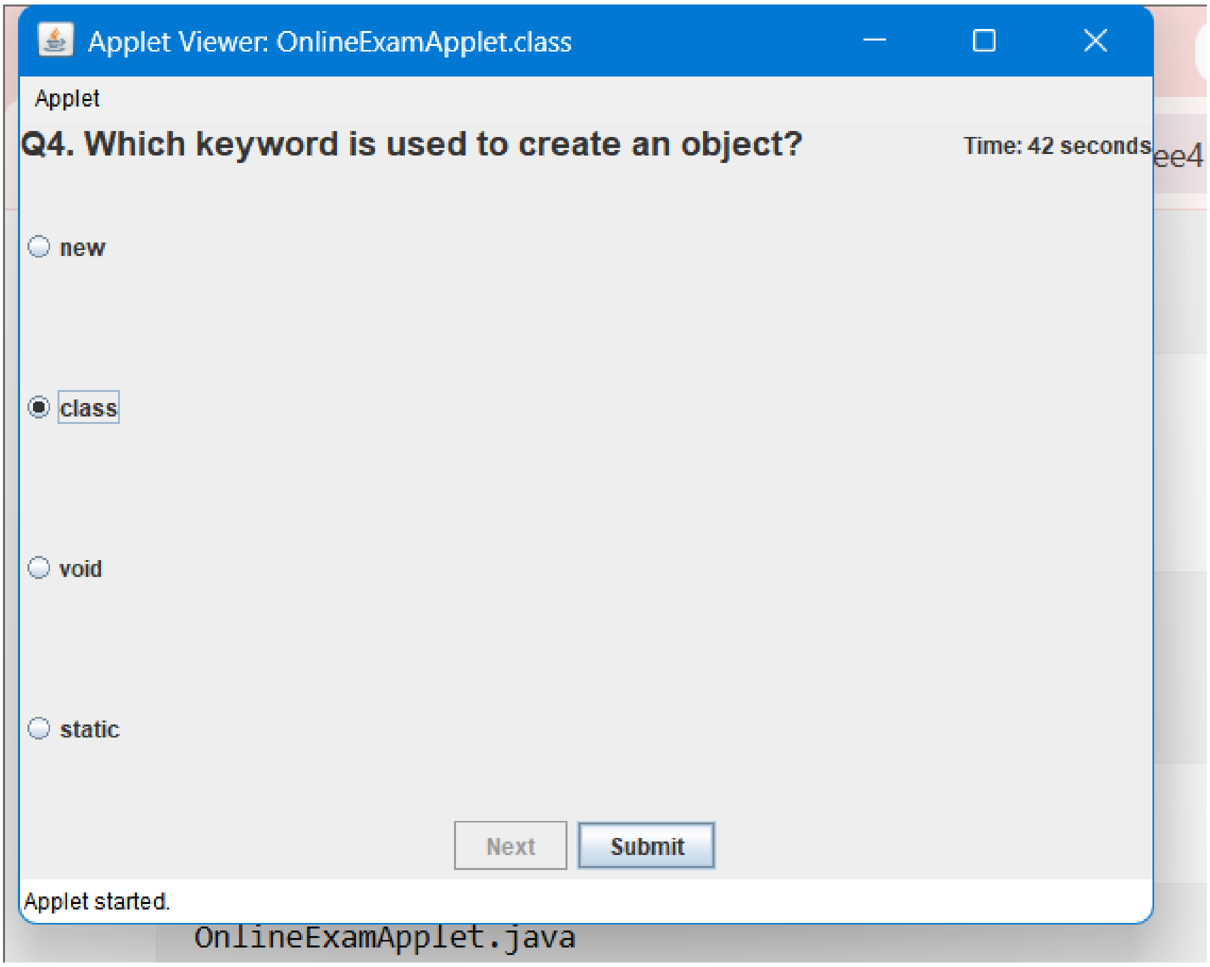
```java
        JMenuItem exitItem = new JMenuItem("Exit");
        startItem.addActionListener(this);
        exitItem.addActionListener(this);
        examMenu.add(startItem);
        examMenu.add(exitItem);
        menuBar.add(examMenu);
        setJMenuBar(menuBar);
        timer = new Timer(1000, new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                time--;
                timerLabel.setText("Time: " + time + " seconds");
                if (time <= 0) {
                    timer.stop();
                    submitExam();
                }
            }
        });

        timer.start();
    }
    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == nextButton) {
            if (!submitted)
                checkAnswer();
                question++;
                if (question == 2) {
                    questionLabel.setText(
                        "Q2. Which keyword is used for inheritance?"
                    );
```

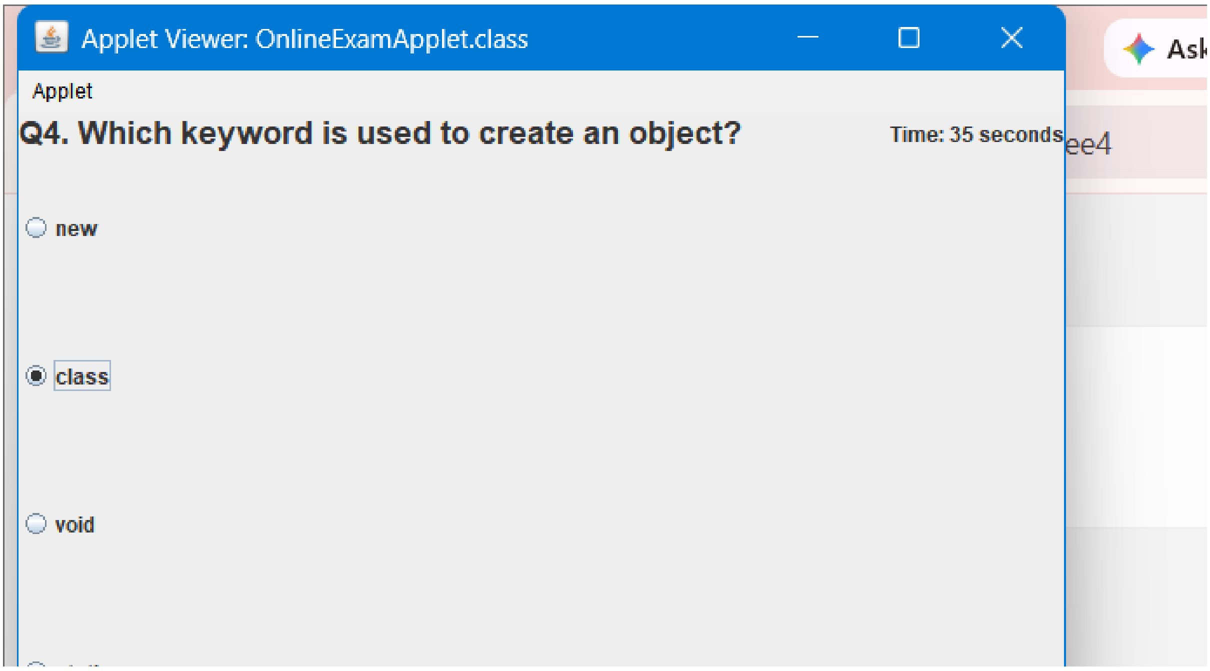
```
            option1.setText("this");
            option2.setText("extends");
            option3.setText("static");
            option4.setText("final");

        }
        else if (question == 3) {

            questionLabel.setText(
                "Q3. Which Swing component is used to create a button?"
            );
            option1.setText("JLabel");
            option2.setText("JPanel");
            option3.setText("JButton");
            option4.setText("JTextField");

        }
        else {
            JOptionPane.showMessageDialog(
                this,
                "All questions completed. Click Submit."
            );
            nextButton.setEnabled(false);
        }
        options.clearSelection();
    }
}
if (e.getSource() == submitButton) {
    submitExam();
```

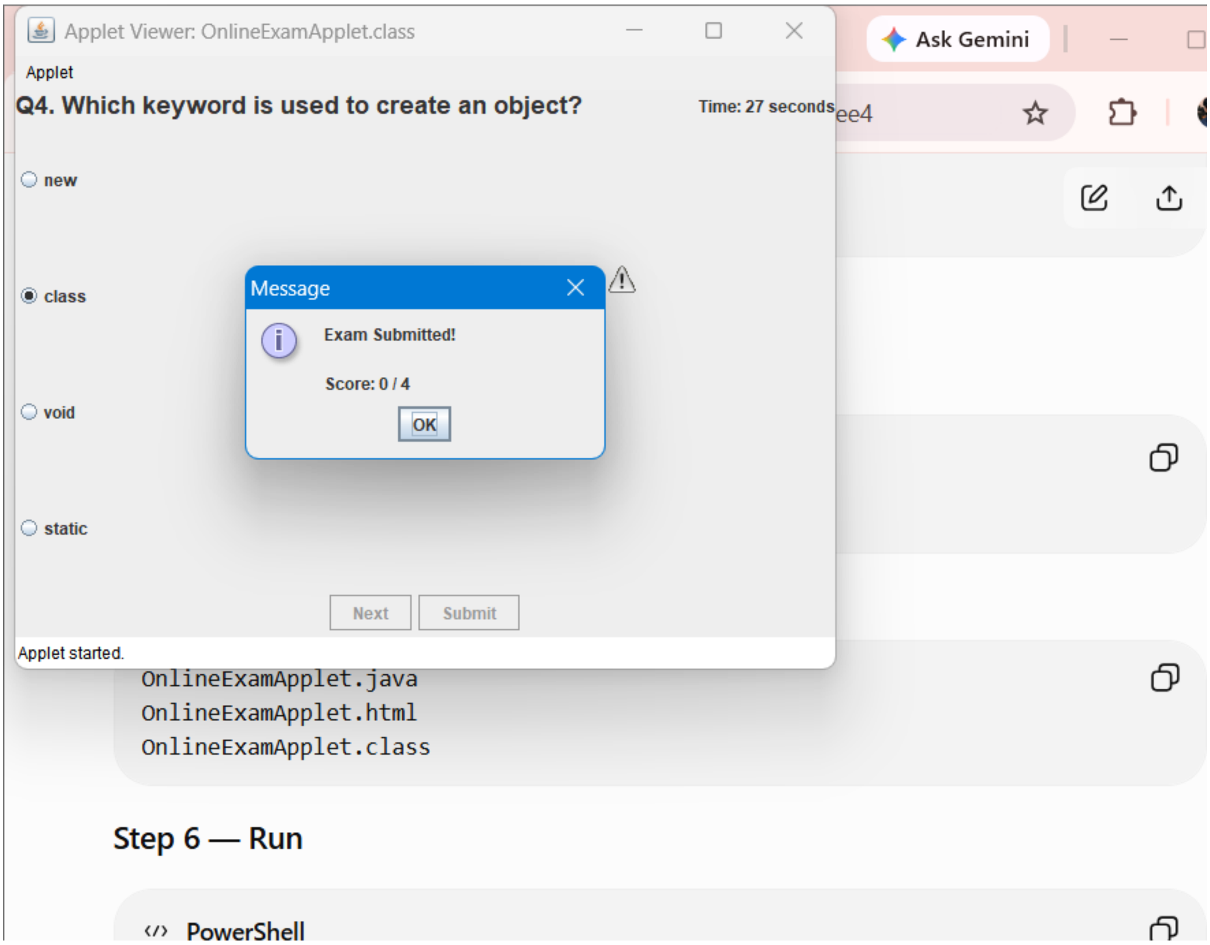
```
    }
    if (e.getActionCommand().equals("Start Exam")) {
        time = 60;
        timer.start();
    }
    if (e.getActionCommand().equals("Exit")) {
        JOptionPane.showMessageDialog(
            this,
            "Exam closed."
        );
    }
}
void checkAnswer() {
    if (question == 1 && option2.isSelected()) {
        score++;
    }
    if (question == 2 && option2.isSelected()) {
        score++;
    }
    if (question == 3 && option3.isSelected()) {
        score++;
    }
}
void submitExam() {
    if (submitted) {
        JOptionPane.showMessageDialog(
            this,
            "Exam already submitted!"
        );
```

```
            return;
        }
        checkAnswer();
        submitted = true;
        timer.stop();
        nextButton.setEnabled(false);
        submitButton.setEnabled(false);
        resultLabel.setText(
            "Score: " + score + " / 3"
        );
        JOptionPane.showMessageDialog(
            this,
            "Exam Submitted!\nYour Score: "
            + score + " / 3"
        );
    }
}
```

OUTPUT SCREEN SHORT:


Applet Viewer: OnlineExamApplet.class
Applet
Q4. Which keyword is used to create an object?
Time: 42 seconds
new
class
void
static
Next
Submit
Applet started.
OnlineExamApplet.java


Applet Viewer: OnlineExamApplet.class
Applet
Q4. Which keyword is used to create an object?
Time: 35 seconds
new
class
void


Applet Viewer: OnlineExamApplet.class
Applet
Q4. Which keyword is used to create an object?
Time: 27 seconds
new
class
void
static
Message
Exam Submitted!
Score: 0 / 4
OK
Next
Submit
Applet started.
Ask Gemini
OnlineExamApplet.java
OnlineExamApplet.html
OnlineExamApplet.class
Step 6 — Run
PowerShell